

1445 ▶ 1545



In 1490, Leonardo da Vinci described capillary action: when water moves up inside a thin tube, in a direction opposed to the force of gravity.

Wooden plate for holding paper

Lever tightens plates together, pressing inked type onto paper.

Solid wooden frame holds plates steady to prevent slippage.

1450

Gutenberg's press

Johannes Gutenberg set up the first European printing press in Mainz, Germany. This used movable type that could be rearranged and reused to make up different pages of text. Producing books was made much easier and the technique spread rapidly throughout Europe.

1489

First use of + and - signs

Johannes Widman, a German mathematician, was the first to use the modern signs for plus (+) and minus (-). Previously, mathematicians had used a variety of signs, including "p" and "m." The sign "=" to mean "equals" came into use later, in 1557.

1464

Trigonometry text

German mathematician Johannes Müller (known by his Latin name, Regiomontanus) wrote *On Triangles*, the first textbook on trigonometry (the study of the relationship between angles and lengths in triangles).



Page from Regiomontanus's *On Triangles*

Replica of Gutenberg's original press



1445

1465

1485

1472

A comet observed

An interest in astronomy led Regiomontanus (Johannes Müller) to make the first detailed observations and descriptions of a comet. Using trigonometrical techniques, he worked out methods for calculating the size of a comet and its distance from Earth.

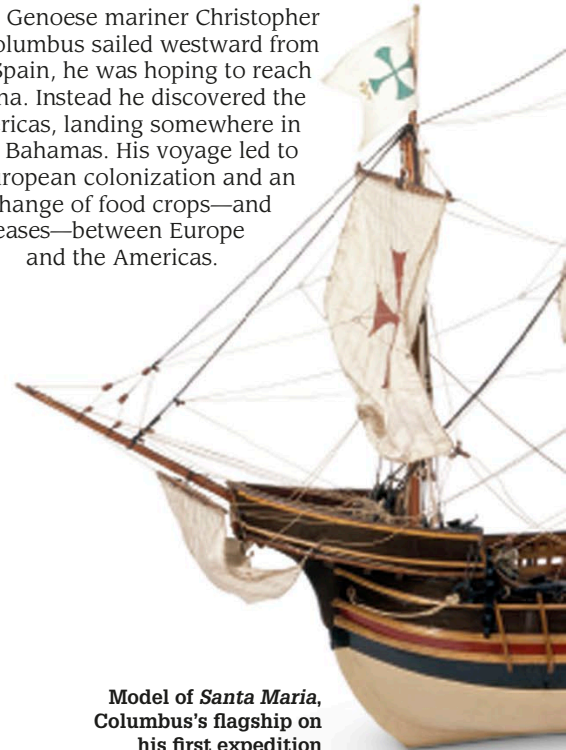


Woodcut of a comet, from the *Nuremberg Chronicle*, 1493

1492

Christopher Columbus discovers America

When Genoese mariner Christopher Columbus sailed westward from Spain, he was hoping to reach China. Instead he discovered the Americas, landing somewhere in the Bahamas. His voyage led to European colonization and an exchange of food crops—and diseases—between Europe and the Americas.



Model of *Santa Maria*, Columbus's flagship on his first expedition



Model of the parachute Leonardo designed in his sketchbooks

Da Vinci's parachute

In his notebooks, Leonardo da Vinci (see pp.58–59) sketched out many ideas for machines centuries before their final invention. In 1481, he drew and described a parachute made of tent canvas.